Deep Updated Subspace Networks for Few-Shot Remote Sensing Scene Classification

Anyong Qin, Fuyang Chen, Qiang Li, Lingyun Tang, Feng Yang, Yue Zhao, and Chenqiang Gao

***Abstract*— Due to the difficulty of manually labeling remote sensing scene images and the demand for the ability to recognize new scene classes, few-shot remote sensing scene classification (FSRSSC) has attracted more and more attention. At present, metric-based FSRSSC methods have made promising progress, especially the prototypical networks-based methods. However, due to the complexity of the background of remote sensing scene images, the prototype classifier, which takes the average features of support samples as the metric benchmark, retains the features of category-irrelevant objects and other background information in the image. This leads to a bad classification result. Therefore, in this work, we propose an FSRSSC method based on the deep updated subspace network (DUSN), which uses class subspace as a metric benchmark to represent the commonality of a category and can effectively mitigate the negative impact of irrelevant objects on the classifier. In addition, for the higher interclass similarity and larger intraclass variance of remote sensing scene images, we further propose an interclass constraint and an intraclass constraint to mitigate the classification confusion. We leverage the interclass constraint to make the images of different classes as far apart as possible, and the intraclass constraint to keep the images of the same class clustered as closely together as possible. Experimental results on three public benchmark datasets demonstrate that our method performs better than the state-of-the-art methods for FSRSSC.**

***Index Terms*— Few-shot learning, metric learning, remote sensing scene classification (RSSC), subspace network.**

I. Introduction

REMOTE sensing scene classification (RSSC) aims to assign a specific class label to each remote sensing scene image based on the understanding of its content [1], so as to distinguish different scenes. It plays an important role in disaster detection [2], [3], urban planning [4], geospatial object detection [5], [6], and land cover classification [7].

Deep learning-based methods which extract image features with convolutional neural networks (CNNs) have gained encouraging results in RSSC [8]. In 2015, Penatti et al. [9] first leveraged CNN for RSSC. After that, a series of CNN-based methods emerged [10], [11], [12], [13], [14], [15], and all showed relatively good classification performance. However, there exist two problems with these methods. First, they rely on a large number of labeled samples for training to obtain a satisfactory model. When the number of annotated remote sensing scene images is small, overfitting [16] will occur, and thus reducing the classification accuracy. Second, they can only classify the trained categories, but cannot recognize new scene categories that often appear in classification tasks.

However, in reality, acquiring a substantial number of labeled remote sensing scene images is extremely challenging, which is labor intensive, requires significant financial resources, and demands professional knowledge. In addition, the openness of the real world leads to the emergence of new unknown scene categories, which have not been seen in training [17], [18]. These factors pose new challenges for RSSC and the above deep learning-based approaches cannot effectively address these challenges.

Under this background, some researchers try to utilize the idea of few-shot learning to solve the RSSC problem [19], [20], [21]. Few-shot learning methods [22], [23], [24], [25], [26], [27] enable the model to learn prior knowledge from base class samples for classifying new class samples and require only a small number of labeled samples. These methods have achieved better classification results in the fields of natural images. But since remote sensing images and natural images are two different types of images, such methods do not show great classification performance on remote sensing images. Compared with natural images, remote sensing scene images have the following characteristics.

A. Complexity of Background

Different from natural images taken from a horizontal view, remote sensing scene images are shot at a higher distance and in an overhead view. This makes the scene images have a large field of view, contain many category-irrelevant ground objects, contain many small-sized target objects, and contain ground objects with huge changes in scale. These reasons lead to a distinct gap between the content of the scene images and the corresponding category. For the first image in Fig. 1, its class

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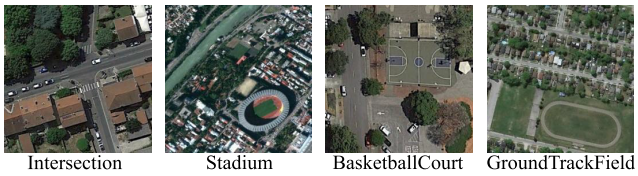

Fig. 1. Complexity of background. (Left–right) Classes of the images are Intersection, Stadium, BasketballCourt, and GroundTrackField.

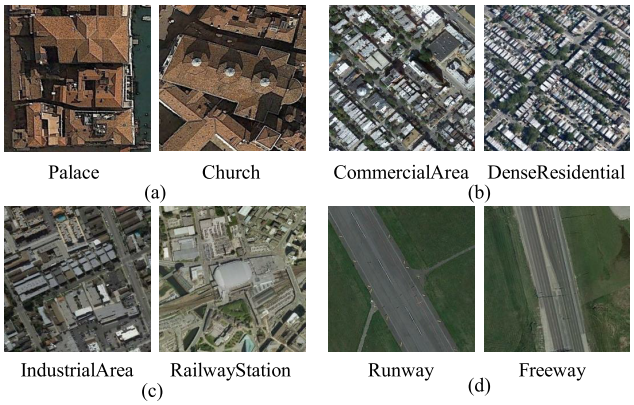

Fig. 2. Higher interclass similarity. (a) Palace and Church. (b) CommercialArea and DenseResidential. (c) IndustrialArea and RailwayStation. (d) Runway and Freeway.

label is the intersection. The target object directly related to the semantic category is the intersection area, but there are also many unrelated objects, such as house buildings, cars, and trees. The other images in Fig. 1 have similar situations.

B. Higher Interclass Similarity

The similarity of objects in shape and texture makes the remote sensing scene images of different classes present a similar appearance, such as the two sets of images, as shown in Fig. 2(a) and (c). In addition, different scene classes may consist of exactly the same ground objects. This is also an important reason for the higher interclass similarity of remote-sensing scene images. For example, in Fig. 2(b), both commercial areas and dense residential consist of buildings, trees, and roads. In Fig. 2(d), both runway and freeway consist of roads and plants.

C. Larger Intraclass Variance

Due to the diversity of ground objects and the differences in illumination, scale, and angle, the scene images in the same category show great visual differences. In Fig. 3, we list three sets of images to visually show the characteristics of remote sensing scene images. Taking Fig. 3(a) as an example, the four images are all churches, but due to the diverse presentations of the church, there are large visual differences between the images.

Because of the above characteristics, few-shot RSSC (FSRSSC) is more challenging. Some researchers have tried to extend the few-shot natural image classification (FSNIC) methods [25], [26], [28] to adapt to the characteristics of remote sensing scene images. Among them, the prototypical networks [25] have gained wide attention. The principle of prototypical networks is to compute class prototypes by averaging the support features of each class, and then classify the

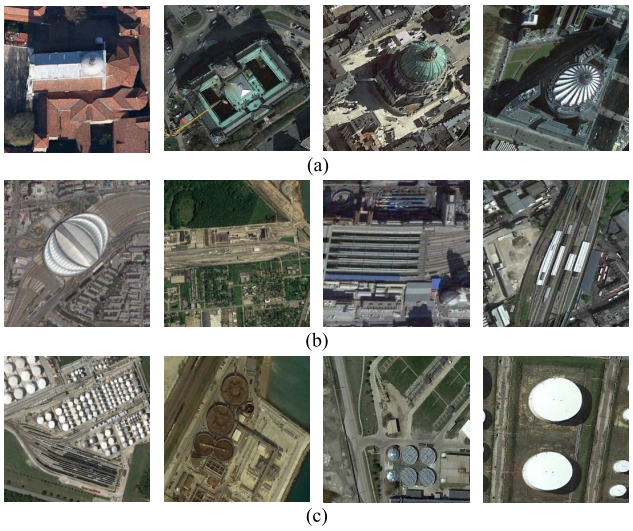

Fig. 3. Larger intraclass variance. (a) Church. (b) RailwayStation. (c) StorageTank.

query images according to the nearest distance between the feature vectors and the prototype vectors. In recent years, many prototypical networks-based FSRSSC methods [29], [30], [31], [32] have been proposed and achieved promising results. For example, self-supervised contrastive learning-based metric learning network (SCL-MLNet) [31] introduced the attention mechanism and self-supervised contrast learning to acquire the representative features of the scene images. Furthermore, more accurate class prototypes can be computed to construct a better prototype classifier. Siamese-prototype network (SPNet) [32] added the prototype self-calibration (SC) and intercalibration (IC) modules to improve the prototypical networks. Although these approaches enhance the reliability of the prototype classifier by obtaining a more accurate prototype representation, they ignore the negative impact of the complex background information of remote sensing scene images on the classifier. Since the scene images contain many ground objects that are not related to the categories, and the prototype classifier will retain these irrelevant object information, which causes great changes to the class prototype. Therefore, the complex background interference will reduce the classification performance of the class prototype.

One of the key steps in the metric-based FSRSSC methods is to choose an appropriate metric benchmark to construct an excellent classifier. The few-shot classifier for RSSC should reduce the influence of irrelevant objects and background information as much as possible. Therefore, in this article, inspired by the subspace learning [33], [34], [35], we propose a subspace classifier, which uses the class subspace as a metric to classify the remote sensing scene images. The subspace can learn the commonality representation of the category by exploiting high-order information embedded in the feature space, thus reducing the complex background interference in building the classifier of scene images. Moreover, to enhance the accuracy of the feature subspace, we introduce the information of the query set to update the class subspaces so that the updated subspace classifier is used for the accurate classification.

In addition, the higher interclass similarity and larger intraclass variance of the remote sensing scene images often cause

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the classification confusion [20]; that is, the scene images of different classes are misclassified into the same class, while the scene images of the same class are classified into different classes. To mitigate the problem of classification confusion, in this work, we add an interclass constraint to keep the scene images of different classes away from each other and an intraclass constraint to make the scene images of the same class clustered.

In summary, the main contributions of this work can be summarized as follows.

1) To mitigate the negative effect of the complex background of remote sensing scene images on the classification, we propose a subspace classifier with class subspace as the metric and further introduce the query set information to obtain more accurate and representative updated subspace. The updated subspace classifier shows better classification performance.
2) We propose an interclass constraint, which increases the distance between class subspaces to make the remote sensing scene images of different classes as far apart as possible.
3) We design an intraclass constraint, which enhances the similarity between each query sample and its refined class prototype, and thus makes the scene images of the same class as close as possible.

The remainder of this article is organized as follows. In Section II, we briefly introduce some related works. Section III describes the proposed method in detail. In Section IV, we present the details of our experiments and the analysis of the results on three public benchmark datasets as well as some ablation experiments. Finally, in Section V, we draw our conclusion.

II. Related Works

In this section, we mainly introduce some research related to our work. The following is a detailed description of the aspects of RSSC, few-shot learning, and FSRSSC.

A. Remote Sensing Scene Classification

RSSC aims to judge which category the input remote sensing scene image belongs to by some algorithm. The early RSSC methods mainly relied on handcrafted or shallow learning-based features, such as bag-of-visual-words (BOVWs) model [36], sparse coding-based features [37], and mid-level visual elements-oriented features [38]. However, with the increase of complexity and resolution of remote sensing scene images, traditional machine learning-based methods are difficult to extract the features and present poor classification performance.

Under this background, deep learning-based approaches get widespread attention. Penatti et al. [9] indicated that pretrained CNN could be used to recognize everyday objects and generalize well to RSSC. This was the first time for CNN to be applied to RSSC. Chaib et al. [10] proposed using discriminant correlation analysis (DCA) to fuse the features extracted by the CNN. The feature fusion method achieved a higher accuracy than solely using the raw deep features. Liu et al. [11] proposed a multiscale CNN (MCNN) framework to solve the problem of the scale variation of objects in remote sensing images. Lu et al. [13] proposed a feature aggregation CNN, which leveraged the information of semantic labels to aggregate the intermediate features for better classification performance. Although deep learning-based methods have achieved excellent performance in RSSC, they often need a large number of annotated data for training. Once the labeled samples are insufficient, overfitting will occur, ultimately leading to poor classification performance. Moreover, these methods can only classify seen categories and cannot be well generalized to new unseen categories.

B. Few-Shot Learning

The core idea of few-shot learning is to learn transferable knowledge from base classes and then leverage it to recognize unlabeled samples from novel classes given only a few labeled samples. The most commonly used few-shot learning methods can be roughly divided into two groups: optimization-based methods and metric-based methods.

The optimization-based methods aim to learn an optimization algorithm that can quickly train an appropriate classifier for novel classes on a few labeled samples. Finn et al. [39] proposed model-agnostic metalearning (MAML) to learn a good set of initialization parameters for the model. For a new classification task, the algorithm can quickly find a set of model parameters suitable for the task by one or very few gradient descents from the learned initialization parameters. Srinivasan et al. [40] proposed an LSTM-based metalearner model (Meta-LSTM) to learn the exact optimization algorithm, which was used to train another learner (such as task-specific classifiers) in the few-shot regime. Li et al. [41] proposed meta-stochastic gradient descent (Meta-SGD), which could quickly adapt a set of learned optimization rules to a learner in only one step. Compared with Meta-LSTM, Meta-SGD is simple, easy to implement and can be learned efficiently. Compared with MAML, in a metalearning process, Meta-SGD not only learns the initialization of the learner but also learns the updated direction and learning rate of the learner, which has greater learning ability.

Metric-based methods utilize the distance between query samples and support samples as a basis for classification. Euclidean distance [25] or cosine distance [42] without parameters and learnable distances [28] with parameters can be used as distance measures. Matching networks (MatchingNet) [43] built different encoders for the support set and query set, and then predicted the labels of query samples according to the distance between samples. Prototypical networks (ProtoNets) [25] calculated a prototype for each category based on the support set and made class prediction according to the distance between query embedding and class prototype. The above methods relied on manually defined similarity metrics for classification. Different from these methods, relation networks (RelationNet) [28] designed a neural network to automatically learn the relational scores among samples to classify the images.

These typical few-shot learning methods mentioned above have been directly applied to FSRSSC. However, these methods do not fully consider the complexity of background, higher interclass similarity, and larger intraclass variance of remote

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sensing scene images, which leads to poor classification performance. Based on this, some researchers attempt to expand the few-shot learning algorithms to adapt to the characteristics of RSSC tasks, so as to obtain a classification model with better performance.

C. Few-Shot Remote Sensing Scene Classification

Over the past few years, some FSRSSC methods have been proposed. Zhai et al. [44] proposed an LLSR model based on metalearning with the ability to learn a classifier from few-shot samples. The knowledge their model learns from one dataset can be easily adapted to a new dataset, which, in turn, would serve in the lifelong few-shot learning. It can be viewed as an optimization-based approach.

In addition, metric-based methods are also widely used in FSRSSC. Li et al. [20] designed an end-to-end network, called discriminative learning of adaptive match network (DLA-MatchNet), which adopted different spatial and channel attention fusion ways to automatically discover the discriminative regions and a nonlinear network to measure the similarity between query set and support set. Li et al. [18] proposed deep metametric learning for FSRSSC, called RS-MetaNet. To maximize the fitting capability and generalization capability, they used a learnable distance metric and a balance loss. Cheng et al. [32] designed an SPNet with prototype SC and IC to acquire more accurate prototypes. Li et al. [31] proposed an SCL-MLNet, which introduced a new attention module to fuse multiscale spatial features from the classification targets of different sizes and used self-supervised contrast learning to extract discriminative features. Cui et al. [45] proposed metakernel networks (MKNs), which adopted a metakernel strategy (MKS) to map the features to the high dimensional space and a new loss function (Stloss) to reduce the dependence of classification boundary on sample selection. Zhang et al. [46] proposed a framework called SGMNet, which carried out similarity measurement at the level of the graph. Gong et al. [47] designed a data architecture network (DANet) from both data and architecture aspects. They introduced a new data augmentation strategy named quad-patch augmentation to enrich the training set and a two-path aggregation attention module to focus on the key clues.

These metric-based methods mentioned above mostly focused on the feature extraction part. They introduced attention mechanisms or multiscale feature learning modules to extract discriminative features of images for classification. However, they neglected the design of the classification metric module (i.e., the classifier). For example, Li et al. [20] adopted a fully connected layer network as the learnable distance metric, which increases the number of model parameters and raises the difficulty of training. Li et al. [31] and Cheng et al. [32] used the prototypical networks-based metric module as the classifier. The prototype classifier they used retains irrelevant object features and other background information in the image, resulting in inaccurate classification, and thus not suitable for FSRSSC. Therefore, in this work, we explore a more suitable few-shot classifier for remote sensing scene images, which uses feature subspace as a class metric and does not have additional learnable parameters.

III. Methodology

In this work, we follow the idea of metric-based few-shot learning and propose an FSRSSC method based on the subspace network, as shown in Fig. 4. We build a subspace classifier with the feature subspace as the metric to mitigate the negative effect of complex background information of remote sensing scene images on the performance of the classifier. And, we add query set features to update the original subspace classifier so that the updated subspace classifier has more accurate class subspaces and better classification ability. In addition, we propose an interclass constraint and an intraclass constraint to mitigate the classification confusion caused by higher interclass similarity and larger intraclass variance of remote sensing scene images.

A. Problem Definition

The goal of FSRSSC is to classify new scene classes with few labeled samples. In this work, we divide each dataset D into the training set D_{train}, validation set D_{val} and test set D_{test} according to the split way described in [40]. Specially, D_{train}, D_{val}, and D_{test} have different and disjoint sets of categories C_{train}, C_{val}, and C_{test}, that is, $C_{\text{train}} \cap C_{\text{val}} = \emptyset$, $C_{\text{train}} \cap C_{\text{test}} = \emptyset$, $C_{\text{val}} \cap C_{\text{test}} = \emptyset$, and $C_{\text{train}} \cup C_{\text{val}} \cup C_{\text{test}} = C_{\text{total}}$.

In this work, we use the episode-based manner of metalearning [48] for training and testing. Therefore, we need to construct a series of tasks. Each task contains N different categories. During task construction, $K + M$ samples per class are selected, among which K samples are used as support sets and the remaining M samples are used as query sets. The task built in this way is called N-way K-shot task. To sum up, each task is composed of support set $S = \{(x_{1,1}, y_{1,1}), (x_{1,2}, y_{1,2}), \ldots, (x_{N,K}, y_{N,K})\}$, and query set $Q = \{q_1, q_2, \ldots, q_{N\times M}\}$, where $x_{i,j}$ denotes the jth sample from class i, $y_{i,j} \in \{0, 1, 2, \ldots, N-1\}$ represents the label of sample $x_{i,j}$, and q_l represents the lth query sample. In particular, the support set and query set do not have the same images in each task.

B. Method Overview

The structure of our framework is shown in Fig. 4. First, a feature extractor based on CNN is used to obtain the feature embeddings of images. We directly flatten the feature map obtained through convolution operations, which is equivalent to mapping the image to a higher dimensional space. Second, we utilize the original prototypes to construct an original subspace classifier. The original subspace classifier is used to estimate the probabilities that query samples belong to each class. Based on these probabilities, we leverage the support set and query set together to refine the original prototypes and construct an updated subspace classifier. Then, we make the final classification according to the updated subspace classifier. In the training process, we adopt an episode-based strategy to train. The model accumulates knowledge from the learning of training tasks to classify novel categories. In addition, for the loss function, we add interclass constraints and intraclass constraints to guide the optimization direction of the feature extractor, aiming to mitigate the problem of classification confusion.

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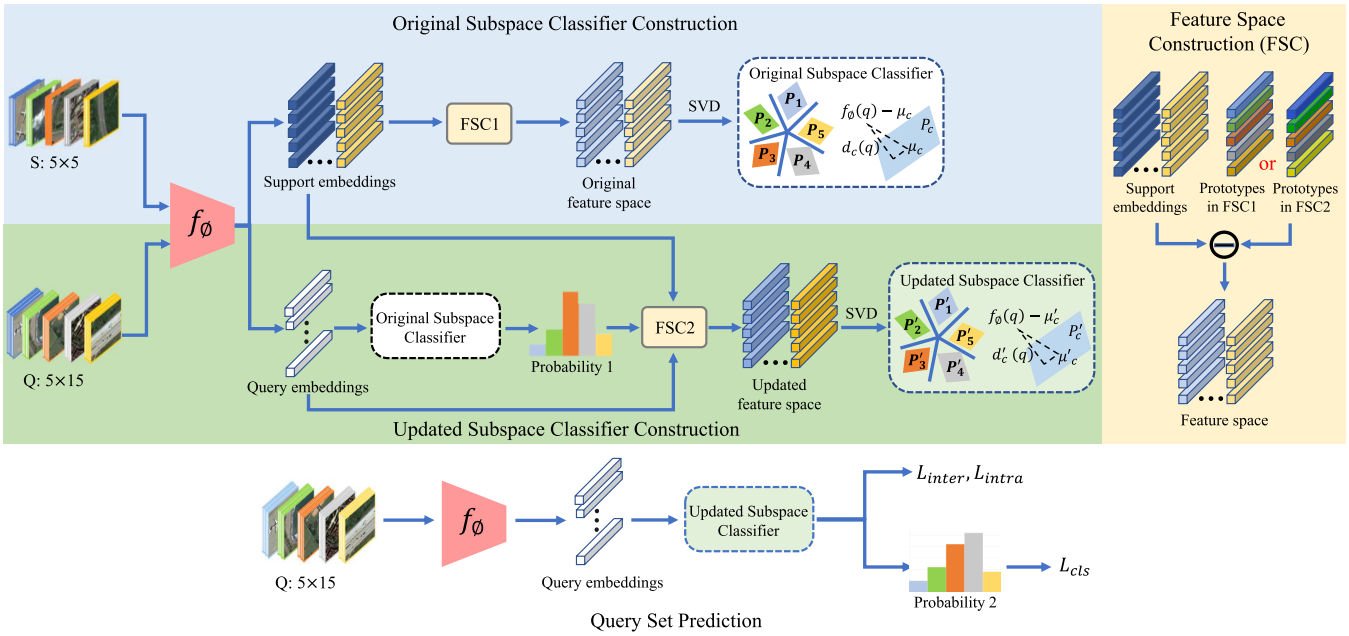

Fig. 4. Overview of our whole framework for a five-way five-shot (S: 5 × 5) setting with 15-query images (Q: 5 × 15). S and Q represent the support set and query set for each task. f_ϕ represents the feature extractor. FSC1 and FSC2 represent the feature space construction modules. FSC1 uses original class prototypes calculated by the mean of support embeddings. FSC2 uses refined class prototypes that combine the support set and query set information. L_{inter} and L_{intra} denote the interclass constraint and intraclass constraint, which aim to keep the interclass images away and the intraclass images clustered, respectively. L_{cls} is the classification loss.

C. Embedding Module

As in [20], we use five convolution blocks (Conv5) as the backbone to learn a feature representation for each remote sensing scene image. Assume that an input image is $x_i \in R^{C\times H\times W}$, where C is the number of input image channels and $H \times W$ is the input image size. Through the above backbone network, we can get a feature map with a specific size $C' \times H' \times W'$. Then, we directly flatten the feature map to a feature vector with length $C'H'W'$. The overall operation is expressed as follows:

$$Z = f_\phi(x_i; \phi) \tag{1}$$

where f_ϕ represents the feature extractor, ϕ is the parameters of the feature extractor, and Z represents the feature vector of the image.

D. Subspace Classifier

In the FSRSSC, besides an appropriate feature extractor, a good classifier is also essential to achieve good classification performance. Most of the current FSRSSC methods use prototypical networks as the classifier. The prototype classifier utilizes the class prototype calculated by averaging the support features as the metric benchmark, which makes the prototype classifier retain much category-irrelevant background information. As a result, the prototype classifier is inaccurate and not suitable for FSRSSC. Therefore, we adopt a subspace classifier for FSRSSC. The subspace classifier leverages class subspace with high-order information as the metric benchmark. The class subspace has a higher feature representation capability than the class prototype. It can represent the commonality of a category, thus reducing the negative effect of the complex background of the remote sensing scene images on the performance of the classifier. Specifically, the construction process of the subspace classifier considers not only the prototype itself but also the relationship between the class prototype and the support set samples.

In this work, in order to obtain a more accurate and suitable classifier for remote sensing scene images, we first build an original subspace classifier to predict the class probabilities of the query set. Then, we introduce the information of the query set to refine the class prototypes. Based on the refined prototypes, we calculate more accurate updated class subspaces to build an updated subspace classifier. The construction processes of the original subspace classifier and updated subspace classifier are described in detail below, respectively.

1) Original Subspace Classifier Construction: In our work, the subspace classifier constructed by the original prototypes is called the original subspace classifier. In N way K shot setting, given an episode with the support set $S = \{(x_i, y_i)\}_{i=1}^{N\times K}$, where x_i is a support image and $y_i \in \{0, 1, 2, \ldots, N-1\}$ is the class label of x_i, we can calculate an original prototype for class c as follows:

$$\mu_c = \frac{1}{K}\sum_{x_j \in S_c} f_\phi(x_j) \tag{2}$$

where $S_c = \{(x_j, y_j)\}_{j=1}^{K}$ represents the support images that belong to class c. From the original class prototype, we conduct the FSC1 in Fig. 4 to obtain the original feature space of each class c, which is expressed as follows:

$$\hat{X}_c = \left[f_\phi(x_{c,1}) - \mu_c, f_\phi(x_{c,2}) - \mu_c, \ldots, f_\phi(x_{c,K}) - \mu_c\right] \tag{3}$$

where μ_c is the original prototype for class c.

By the singular value decomposition (SVD) of $\hat{X}_c$, we can obtain a matrix B_c with an orthogonal basis ($B_c B_c^{\mathrm{T}} = I$).

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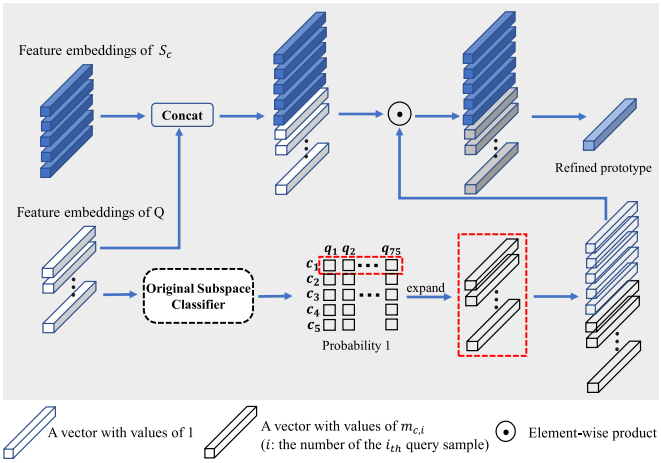

Fig. 5. Prototype refining method. We combine query set features with support set features to obtain the refined prototype for each class c.

We treat the truncation matrix of B_c as the class-specific projection matrix, which also represents the subspace of class c, denoted by P_c. We use the class-specific projection matrix to calculate the projection distance of the query sample to each class subspace. The core idea of a subspace classifier is to find the closest distance between a data point and its projection on the subspace. For a query sample q_i, we first project it onto each P_c, and then classify it according to the projected distance. The projected distance is defined as follows:

$$d_c(q_i) = ||(I - M_c)\big(f_\phi(q_i) - \mu_c\big)||_2 \tag{4}$$

where $M_c = P_c P_c^{\mathrm{T}}$, μ_c is the original prototype for class c and $||.||_2$ represents Euclidean norm ($L2$ Norm). For a vector $e = [e_1, e_2, \ldots, e_n]$, the $L2$ norm is calculated as $||e||_2 = (e_1^2 + e_2^2 + \cdots + e_n^2)^{1/2}$. According to the above-projected distance, we use a softmax function to predict the probability that query sample q_i belongs to class c, which can be written as follows:

$$p(y = c|q_i) = \frac{\exp(-d_c(q_i))}{\sum_{c'} \exp(-d_{c'}(q_i))} \tag{5}$$

where $c' \in \{0, 1, 2, \ldots, N-1\}$ represents the class in each task.

As aforementioned, we utilize the relationship between support set features and class prototype to represent the feature space of a class and then compute the class subspace as a class metric for the classifier construction. A more accurate class prototype helps to obtain a class subspace that more closely matches the class features. Therefore, we add the image features of the query set to refine the prototypes and use the refined prototypes to get updated class subspaces for the construction of an updated subspace classifier.

2) Updated Subspace Classifier Construction: In our work, the subspace classifier constructed by the refined prototypes is called an updated subspace classifier. First, we obtain the refined prototypes according to the prototype refining method, as shown in Fig. 5.

For each query sample q_i, we use the original subspace classifier to estimate the probability that it belongs to class c as follows:

$$m_{c,i} = \frac{\exp(-d_c(q_i))}{\sum_{c'} \exp(-d_{c'}(q_i))} \tag{6}$$

where $c' \in \{0, 1, 2, \ldots, N-1\}$ represents the class in each task.

Then, we leverage the category prediction probabilities of the query samples to refine the prototype according to

$$\mu_c' = \frac{\sum_{x_j \in S_c} f_\phi(x_j) + \sum_i m_{c,i} f_\phi(q_i)}{K + \sum_i m_{c,i}} \tag{7}$$

where μ_c' is the refined prototype for class c.

Based on the refined prototype, we conduct the FSC2 in Fig. 4 to obtain an updated feature space for each class c, which is expressed as follows:

$$\hat{X}_c' = \big[f_\phi(x_{c,1}) - \mu_c', f_\phi(x_{c,2}) - \mu_c', \ldots, f_\phi(x_{c,K}) - \mu_c'\big]. \tag{8}$$

In the same way, SVD is carried out, and the new projected matrix P_c' of each class is obtained. With these steps, we build an updated class subspace for each category. The projected distance from the query sample to the updated class subspace is used for the classification prediction of query samples. The new projected distance is calculated as follows:

$$d_c'(q_i) = ||\big(I - M_c'\big)\big(f_\phi(q_i) - \mu_c'\big)||_2 \tag{9}$$

where $M_c' = P_c' P_c'^T$ and $||.||_2$ represent the Euclidean norm ($L2$ norm). We adopt the softmax function to calculate the probability that q_i belongs to class c

$$p'(y = c|q_i) = \frac{\exp\big(-d_c'(q_i)\big)}{\sum_{c'} \exp\big(-d_{c'}'(q_i)\big)}. \tag{10}$$

Finally, we classify the query sample into the class corresponding to the maximum prediction probability.

E. Loss Function

To mitigate the problem of classification confusion, we add an interclass constraint to keep the scene images of different classes away from each other and an intraclass constraint to make the scene images of the same class clustered. Therefore, our loss function consists of three parts, which are used to optimize the model parameters in the training process.

In the first part, starting from the task itself, we use the cross-entropy loss function to calculate the classification loss of the FSRSSC task, so as to make as many samples as possible be correctly classified

$$L_{\mathrm{cls}} = -\frac{1}{NM} \sum_q \log\big(p_{c,q}\big) \tag{11}$$

where N indicates the number of classes in a task, M indicates the number of query samples per class, and $p_{c,q}$ represents the predicted probability that the query sample q belongs to its true class c.

In the second part, we address the problem of higher interclass similarity in remote sensing scene images by enlarging the interclass distance. Specifically, given any two class subspaces P_i and P_j, the distance metric between these two subspaces is defined as follows:

$$d_p^2\big(P_i, P_j\big) = ||P_i P_i^T - P_j P_j^T||_F^2 = 2n - 2||P_i^T P_j||_F^2. \tag{12}$$

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Our goal is to maximize $d_p^2(P_i, P_j)$, which is the same as minimizing $||P_i^T P_j||_F^2$. Therefore, we express the interclass constraint as follows:

$$L_{\text{inter}} = \frac{2}{N(N-1)} \sum_{i=1}^{N} \sum_{\substack{j=1 \\ j>i}}^{N} ||P_i^T P_j||_F^2 \tag{13}$$

where N is the number of classes in each task.

Finally, we reduce the intraclass variance of the remote sensing scene images by maximizing the similarity between each query sample and the refined prototype of its real class, so that the feature vectors of samples of the same category have a greater similarity. It can also be understood that we use the refined prototype as the reference standard for each category. We enhance the similarity between the image feature vector and the refined prototype to make the feature extractor pay more attention to the similar information within the class. The intraclass constraint is defined as follows:

$$L_{\text{intra}} = 1 - \frac{1}{NM} \sum_{q} \text{cosine}\langle q, \mu_c' \rangle \tag{14}$$

where $\text{cosine}\langle , \rangle$ denotes the cosine similarity, and μ_c' is the refined prototype corresponding to the real category of the query sample q.

In summary, our loss function is expressed as follows:

$$L = L_{\text{cls}} + \alpha L_{\text{inter}} + \beta L_{\text{intra}} \tag{15}$$

where α and β are the hyperparameters that control the weights of L_{inter} and L_{intra}. Our extensive experiments show that the model achieves the best performance when $\alpha = 0.3$ and $\beta = 0.7$.

IV. Experiments

To demonstrate the effectiveness of our method, we conduct a series of experiments on three public datasets. First, three benchmark datasets of RSSC and some implementation details of the experiment are briefly introduced. Then, we discuss the experimental results and compare them with other methods. Finally, ablation experiments are carried out to prove the validity of the key parts of our works.

A. Dataset Description

In the experiments, we use three public datasets to evaluate the performance of our method, including NWPU-RESISC45 dataset [49], UC Merced dataset [36], and AID dataset [50]. To ensure the fairness and rationality of the comparison with other methods, we split the training set, validation set, and test set in the same way as in [20], [32], and [46]. The training set is mainly used to train the model, the validation set is used to select the optimal model for saving, and the test set is used to evaluate the overall performance of our method. The detailed splits of each dataset are shown in Table I.

The NWPU-RESISC45 dataset is an available large-scale dataset for RSSC published by Northwestern Polytechnical University (NWPU) in 2017. The dataset consists of 31 500 RGB images covering 45 scene categories, each of which has 700 RGB images with a size of 256 × 256 pixels. For most scene categories, the change in spatial resolution is approximately 0.2–30 m per pixel. For the NWPU-RESISC45 dataset, the number of classes for the training set, validation set, and test set is 25, 10, and 10, respectively (see Table I for more details).

The UC Merced dataset was created by the UC Merced Computer Vision Laboratory in 2010. It contains a total of 2100 RGB images from 21 categories and is widely used in land use classification research. There are 100 RGB images with a size of 256 × 256 pixels per class and a spatial resolution of 0.3 m per pixel. For the UC Merced dataset, we split it into ten categories for the training set, five categories for the validation set, and six categories for the test set, respectively (see Table I for more details).

The AID dataset was jointly proposed by the Huazhong University of Science and Technology and Wuhan University in 2017. The dataset has 10 000 RGB images of 30 different scene classes and about 220–420 images of size 600 × 600 pixels in each class. The spatial resolution ranges from approximately 0.5-8m. For the AID dataset, we choose 16 classes as the training set, seven classes as the validation set, and the remaining seven classes as the test set (see Table I for more details).

B. Implementation Details

We implement all the experiments in PyTorch [51] framework with CUDA 11.4 using a single NVIDIA GeForce RTX 3090 with 24-GB memory. We utilize Conv5 as the feature extractor. Each convolutional block is composed of a 3 × 3 convolution kernel layer, a normalization layer, a rectified linear unit (ReLU) activation nonlinear layer, and a 2 × 2 max-pooling layer. The output channels of each convolution block are 32, 64, 128, 256, and 256 in turn. For each dataset, we resize all remote sensing scene images to 128 × 128 as the input of the feature extractor and then train the model from scratch without using pretrained weights. In this work, we use the adaptive momentum estimation algorithm (ADAM) [52] to optimize the feature extractor. The initial learning rate is set as 0.001 and the learning rate is updated with the increase of the training epoch.

In our experiments, we adopt an episode-based strategy to train. Each episode consists of a support set and a query set, both of which have five classes. According to different experimental settings, the number of support images selected for each category is different. In the case of five-way one-shot, there is only one support image for each class. However, according to the design of our method, we need more than one sample to get the subspace of each class. Therefore, in the case of one-shot, we flip the only one support image to generate another image. In the case of a five-way five-shot, five samples per class are randomly selected as support images. In addition, 15 samples different from the support set are randomly selected for each category as the query set. Finally, we use the overall accuracy to evaluate the performance of our approach, which is defined as the proportion of correctly classified samples to the total samples. We report the average accuracy over 600 randomly generated tasks from the test set on five-way one-shot and five-way five-shot with 95% confidence intervals for all our experiments, which can be

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TABLE I
DETAILED SPLITS ON THREE DATASETS

Datasets	Training Set	Validation Set	Test Set
NWPU-RESISC45	Airplane; BaseballDiamond; Beach; Bridge; Chaparral; Church; Cloud; Desert; Freeway; GolfCourse; Harbor; Island; Lake; Meadow; MobileHomePark; Mountain; Palace; Railway; RectangularFarmland; Roundabout; SeaIce; Ship; SparseResidential; Stadium; Wetland	ThermalPowerStation; Overpass; RailwayStation; TennisCourt; CommericalArea; Snowberg; StorgeTank; Runway; Terrace; IndustrialArea	CircularFarmland; ParkingLot; Forest; DenseResidential; River; GroundTrackField; Intersection; MediumResidential; Airport; BasketballCourt
UC Merced	Agricultural; BaseballDiamond; Buildings; Chaparral; DenseResidential; Freeway; Harbor; MediumResidential; Overpass; ParkingLot	Airplane; Forest; Intersection; Runway; StorageTanks	SparseResidential; GolfCourse; MobileHomePark; TennisCourt; River; Beach
AID	Bench; Bridge; Center; Commerical; Desert; Port; Farmland; Mountain; Park; Parking; Playground; RailwayStation; Resort; Square; Stadium; StorageTanks	BareLand; BaseballField; Church; Meadow; Pond; River; School	Airport; DenseResidential; Forest; Industrial; MediumResidential; SparseResidential; Viaduct

expressed as follows:

$$\text{Acc} = \frac{1}{T_{ts}} \sum_{i=1}^{T_{ts}} \frac{r^{(i)}}{NM} \tag{16}$$

where $r^{(i)}$ is the number of samples that are correctly classified in the query set of task i, N represents the number of categories in each task, M represents the number of query images per class, and T_{ts} is the total number of tasks used in the evaluation.

C. Experimental Results and Comparisons

To verify the effectiveness of our method, we choose several typical few-shot learning methods to compare with ours. These peer competitors can be divided into two groups: optimistic-based methods such as MAML [39] and Meta-SGD [41], and metric-based methods such as MatchingNet [43], ProtoNet [25], and RelationNet [28]. In addition, we also compare with recent methods for FSNIC and FSRSSC, including DeepEMD [26], FRN [27], MCL-Katz [53], GLIML [54], ISC-LD [55], LLSR [44], RS-MetaNet [18], DLA-MatchNet [20], SGMNet [46], SPNet [32], and SCL-MLNet [31]. Below we will illustrate the experimental results of our method and the comparison methods on three public datasets under five-way one-shot and five-way five-shot task settings. For MAML, Meta-SGD, MatchingNet, ProtoNet, RelationNet, DeepEMD, LLSR, RS-MetaNet, DLA-MatchNet, and SGMNet, we adopt Conv5 as the backbone which is consistent with ours and refer to the experimental results in [46]. For FRN, MCL-Katz, GLIML, and ISC-LD, the experimental results are implemented by ourselves. For SPNet and SCL-MLNet, we use the backbone and experimental results the same as those in their original papers.

Table II shows the results of the experiments on the NWPU-RESISC45 dataset. We can see that our method achieves 73.84% (one-shot) and 84.96% (five-shot) in the overall accuracy, respectively. It indicates that our method's classification performance is superior to all other comparisons. In the prototype-based methods, our approach has a significant improvement over ProtoNet in the overall accuracy of one-shot and five-shot scenarios. In particular, in the case of one-shot, the overall classification accuracy of our method is 8.84% higher than SPNet and 18.69% higher than SCL-MLNet. In the case of five-shot, our method is almost 1.22% and 5.07% more accurate than SPNet and SCL-MLNet.

TABLE II
FEW-SHOT CLASSIFICATION ACCURACY (%) WITH 95% CONFIDENCE INTERVALS OF DIFFERENT METHODS ON NWPU-RESISC45 DATASET

Method	NWPU-RESISC45	
	1-shot	5-shot
MAML [39]	48.40±0.82	62.90±0.69
MetaSGD [41]	60.63±0.90	75.75±0.65
LLSR [44]	51.43	72.90
ProtoNet [25]	40.33±0.18	63.82±0.56
MatchingNet [43]	37.61	47.10
RelationNet [28]	66.43±0.73	78.35±0.51
DeepEMD [26]	64.39±0.84	78.01±0.56
FRN [27]	64.98±0.42	81.65±0.25
MCL-Katz [53]	63.30	80.78
GLIML [54]	66.86±0.68	78.91±0.45
ISC-LD [55]	69.26±0.67	82.70±0.39
DLA-MatchNet [20]	68.80±0.70	81.63±0.46
RS-MetaNet [18]	64.07±0.90	79.62±0.65
SGMNet [46]	70.40±0.83	82.32±0.53
SPNet [32]	67.84±0.87	83.94±0.50
SCL-MLNet [31]	62.21±1.12	80.86±0.76
Ours	**73.84±0.91**	**84.96±0.46**

For the UC Merced dataset, as shown in Table III, our method achieves the best results under both one-shot and five-shot scenarios. The overall accuracy of our method in five-way five-shot scenario is 79.44%, which is 8.05% higher than the second-best method SPNet. In the case of one-shot, our method outperforms the second-best method SGMNet by a low margin of 2.78%.

In addition, we also conduct experiments on the AID dataset, and the experimental results are presented in Table IV. Since no experiment on the AID dataset was conducted in [32], it is not included in our study. From the comparison of other methods, it can be found that we achieve state-of-the-art results on five-way one-shot and five-way five-shot. In detail, in the case of five-shot, the classification accuracy of our method is higher than any other method with a value of 83.36%, which is 2.18% higher than ISC-LD. When only given one labeled sample, our method surpasses the top 2 approach by 4.31%.

In summary, on the three public datasets, our method achieves the best results compared with other methods. By the analysis of experimental results, we can conclude that our method is more effective and suitable for FSRSSC. Compared with the class prototype, the class subspace can accurately represent the commonality of a category and reduce the interference of category-irrelevant background information in

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TABLE III

FEW-SHOT CLASSIFICATION ACCURACY (%) WITH 95% CONFIDENCE INTERVALS OF DIFFERENT METHODS ON UC MERCED DATASET

Method	UC Merced	
	1-shot	5-shot
MAML [39]	48.86±0.74	60.78±0.62
MetaSGD [41]	50.52±2.61	60.82±2.00
LLSR [44]	39.47	57.40
ProtoNet [25]	52.27±0.20	69.86±0.15
MatchingNet [43]	34.70	52.71
RelationNet [28]	48.08±1.67	61.88±0.50
DeepEMD [26]	58.47±0.76	70.42±0.58
FRN [27]	50.89±0.37	68.34±0.30
MCL-Katz [53]	50.73	68.95
GLIML [54]	56.41±0.62	70.40±0.41
ISC-LD [55]	56.50±0.58	70.66±0.43
DLA-MatchNet [20]	53.76±0.62	63.01±0.51
RS-MetaNet [18]	49.68±0.71	67.53±0.59
SGMNet [46]	60.52±0.74	73.42±0.49
SPNet [32]	57.64±0.73	73.52±0.51
SCL-MLNet [31]	51.37±0.79	68.09±0.92
Ours	**62.20±0.84**	**79.44±0.47**

TABLE IV

FEW-SHOT CLASSIFICATION ACCURACY (%) WITH 95% CONFIDENCE INTERVALS OF DIFFERENT METHODS ON AID DATASET

Method	AID	
	1-shot	5-shot
MAML [39]	43.20±0.77	60.37±0.75
MetaSGD [41]	45.01±0.98	62.58±0.80
LLSR [44]	45.18	61.76
ProtoNet [25]	54.32±0.86	67.80±0.64
MatchingNet [43]	33.87	50.40
RelationNet [28]	54.62±0.80	68.80±0.66
DeepEMD [26]	61.04±0.77	74.51±0.55
FRN [27]	62.29±0.37	79.33±0.24
MCL-Katz [53]	55.28	75.66
GLIML [54]	61.28±0.61	79.56±0.41
ISC-LD [55]	63.27±0.59	81.58±0.37
DLA-MatchNet [20]	61.99±0.94	75.03±0.67
RS-MetaNet [18]	58.51±0.84	73.76±0.69
SGMNet [46]	62.21±0.81	75.68±0.56
SPNet [32]	-	-
SCL-MLNet [31]	59.46±0.96	76.31±0.68
Ours	**66.00±0.86**	**83.36±0.46**

remote sensing scene images. And, our loss function can increase the interclass distance and focus on the intraclass similar information to help obtain the best model. In addition, during the test phase, our approach does not require fine-tuning the model parameters while the test task changes, showing the proposed method is simpler and more efficient.

D. Selection of Hyperparameters

1) Weighting Coefficients of the Loss Function: Our model is trained under the joint supervision of few-shot classification loss, interclass constraint, and intraclass constraint. In our experiments, we set the weighting coefficient of few-shot classification loss as 1 and the sum of the weighting coefficients of interclass constraint and intraclass constraint as 1. To obtain the optimal loss function, we conduct several experiments to select the most appropriate weighting coefficients for interclass and intraclass constraints. Fig. 6 shows the results of our experiments.

The x-axis of the line graph represents the ratio of interclass constraint to intraclass constraint, namely, α/β. Since the sum of α and β is a fixed value of 1, we divide them with equal intervals of 0.1. There are altogether nine sets of values for the weighting coefficients α/β. The y-axis of the line graph represents the average classification accuracy of 600 random test tasks. To ensure the fairness of comparison, we set a fixed random seed to ensure that the same training tasks and test tasks are selected for each experiment. The solid line in Fig. 6 represents the comparison results of the three datasets in the case of five-shot, while the dashed line represents the experimental results of the three datasets in the case of one-shot. From the line chart, as α increases from 0.1 to 0.3, there is an increasing trend for all lines except for the case of one-shot on NWPU-RESISC45. In addition, when α is between 0.6 and 0.9, the classification accuracy tends to decrease in the case of five-shot on UC Merced, increase and then stabilize in the case of one-shot on AID and remain stable in the other cases. Specially, when α/β is 0.3/0.7, the classification accuracies of the three datasets all achieve the highest under one-shot and five-shot scenarios. Therefore, we set the weighting coefficients of interclass constraint and intraclass constraint as 0.3/0.7.

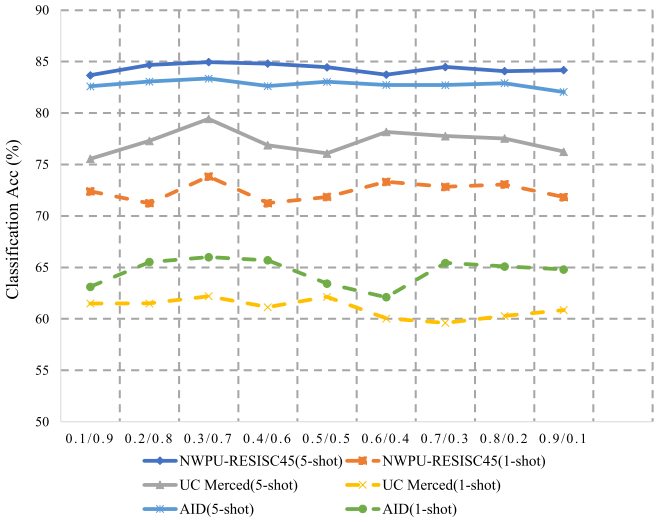

Fig. 6. Comparison of different weighting coefficient settings for interclass and intraclass constraints on three public datasets.

2) Dimensionality of the Subspace: Our method utilizes the feature subspace as the metric to construct a subspace classifier. Different subspace dimensionalities (n) generate different feature subspaces, which may influence the classification performance. Therefore, we conduct experiments on the NWPU-RESISC45, UC Merced, and AID datasets to help to select an appropriate subspace dimensionality. In the case of five-shot, according to the number of support samples, we choose n from 1 to 5 for the experiment. The experimental results are shown in Fig. 7. As n changes from 1 to 5, the few-shot classification accuracy first increases and then decreases. This may be the reason that when the subspace dimensionality is too small, the feature subspace lacks the key features, and when the subspace dimensionality is 5, the redundant information is introduced into the subspace. Therefore, in the case of five-shot, we choose the subspace dimensionality as 4. In the case of one-shot, since only one labeled real scene image is used, we set the subspace dimensionality as 1.

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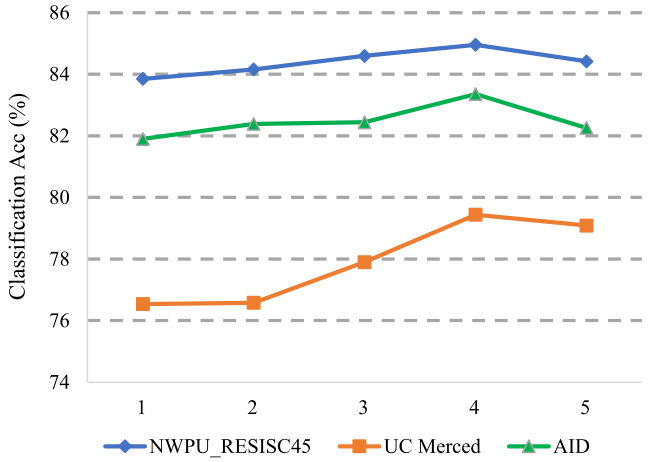

Fig. 7. Comparison of the classification performance (five-shot) with different subspace dimensionalities on three public datasets.

E. Ablation Study

In this section, we first assess the performance of different methods on the test sets with varying background complexities to show that our proposed method can mitigate the negative impact of complex backgrounds on the classifier. Then, we investigate the effectiveness of an updated subspace classifier and the influence of different prototype refining methods, feature extraction networks, and loss function settings on classification performance.

1) Impact of Image Background Complexity on the Performance of Different Methods: Since the NWPU-RESISC45 dataset contains the most scene images, the most scene categories, and more complex and variable images [20], [45], so the NWPU-RESISC45 dataset is used to construct three test sets with varying background complexities: original test, complex test, and simple test. A detailed description of each test set is given below.

Original Test: It consists of all the images from the test set of the NWPU-RESISC45 dataset, with ten categories and 700 images per category. Both the images with complex backgrounds and images with simple backgrounds are included.

Complex Test: It contains a selection of images with complex backgrounds from the test set of the NWPU-RESISC45 dataset. There are ten categories and 100 images in each category.

Simple Test: There are also 10 categories in total. Each category consists of 100 images with simple backgrounds from the test set of the NWPU-RESISC45 dataset. Fig. 8 displays the simple and complex background images of the same classes. As can be seen from Fig. 8, the images from the simple test contain fewer irrelevant objects. Moreover, the target objects related to the category cover a significant area within the entire image. However, the images from the complex test data contain more irrelevant objects, and the target objects associated with the category are relatively smaller in size.

Then, we test the performance of the prototype classifier-based method, original subspace classifier-based method and updated subspace classifier-based method on the three test sets. The experimental results are shown in Tables V and VI.

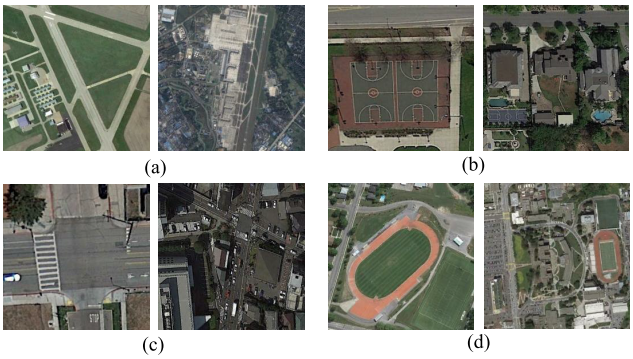

Fig. 8. Images with simple/complex backgrounds. In (a)–(d), the left image is a simple background image from simple test and the right image is a complex background image from a complex test. (a) Airport. (b) BasketballCourt. (c) Intersection. (d) GroundTrackField.

From Tables V and VI, we can find the following.

1) No matter which classifier-based methods, they achieve the highest classification accuracy on simple test and the lowest classification accuracy on complex test. This shows that the complex background has a large impact on FSRSSC. A classifier that can abstract the commonality of each category and mitigate the negative influence of complex backgrounds is essential to improve the accuracy of FSRSSC.
2) On the three test sets, the subspace classifier-based method achieves better classification results on both one-shot and five-shot compared with the prototype classifier-based method. In particular, for the complex test: at one-shot, the classification accuracies of the original subspace classifier-based method and our proposed updated subspace classifier-based method reach 62.21% and 69.76%, respectively, which surpass the prototype classifier-based method by 6.31% and 19.21%. At five-shot, the classification accuracies of the original subspace classifier-based method and our proposed updated subspace classifier-based method are 78.25% and 82.40%, respectively, which are 3.61% and 9.11% higher than the prototype classifier-based method.

The experimental data illustrate that even on complex background images, our proposed updated subspace classifier-based method still shows better classification performance than the prototype classifier-based method. This also just shows that our proposed updated subspace classifier-based method is less affected by the complex background of remote sensing scene images. Therefore, our proposed updated subspace classifier-based method can mitigate the negative influence of the complex background of remote sensing images on the classifier and achieve better performance.

2) Effectiveness of Updated Subspace Classifier: In this part, we study whether updating the subspace classifier is helpful in improving classification performance. We set up two groups of experiments. One group does not update the subspace classifier; that is, the original subspace classifier is directly used for the final classification. The other group updates the subspace classifier; that is, the updated subspace classifier is used for the final classification. Table VII shows the results of our experiments on NWPU-RESISC45, UC Merced, and AID datasets under one-shot and five-shot

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TABLE V

FEW-SHOT (ONE-SHOT) CLASSIFICATION ACCURACY (%) WITH 95% CONFIDENCE INTERVALS OF DIFFERENT METHODS ON ORIGINAL TEST, COMPLEX TEST, AND SIMPLE TEST OF THE NWPU-RESISC45 DATASET

Method	1-shot		
	Original Test	Complex Test	Simple Test
prototype classifier-based method	60.46±0.88	58.52±0.86	70.84±0.84
original subspace classifier-based method	64.94±0.88	62.21±0.92	76.39±0.80
updated subspace classifier-based method	73.84±0.91	69.76±0.96	84.79±0.73

TABLE VI

FEW-SHOT (FIVE-SHOT) CLASSIFICATION ACCURACY (%) WITH 95% CONFIDENCE INTERVALS OF DIFFERENT METHODS ON ORIGINAL TEST, COMPLEX TEST, AND SIMPLE TEST OF THE NWPU-RESISC45 DATASET

Method	5-shot		
	Original Test	Complex Test	Simple Test
prototype classifier-based method	78.73±0.56	75.52±0.67	87.41±0.49
original subspace classifier-based method	81.56±0.53	78.25±0.65	90.66±0.41
updated subspace classifier-based method	84.96±0.46	82.40±0.57	93.27±0.34

conditions. We can see that the network using the updated subspace classifier achieves higher classification accuracy in the cases of one-shot and five-shot. Especially, in the case of a five-way one-shot, compared with the original subspace classifier without updating, the overall accuracy of the updated subspace classifier is greatly improved, with a maximum improvement of 13.70% on the NWPU-RESISC45 dataset.

The experimental results strongly demonstrate the effectiveness of updating the subspace classifier. This is because when constructing the updated subspace classifier, we use the refined prototype. Compared with the original prototype, the refined prototype introduces the information of the query set for each class, which can more comprehensively reflect the feature of a class. Notably, in the case of five-way one-shot, the original subspace classifier utilizes only one support sample to get the original prototype, which has large randomness and low validity. While our updated subspace classifier uses the refined class prototype, which is calculated by more samples. As a result, the refined prototype is more accurate than the original prototype. Therefore, the class subspace obtained by the refined prototype can more accurately represent the commonality of a category. The updated subspace classifier has better classification performance.

3) Influence of Different Prototype Refining Methods: In this work, we introduce the query set to refine the original prototype. The key step is to obtain the probabilities that query samples belong to each class. Different calculation methods of class probability for query sets determine different prototype refining methods. Next, we explore the influence of different prototype refining methods. Most of the prototype refining methods are based on the prototype distance, i.e., the probabilities that query samples belong to each category are calculated according to the distance between query samples and class prototypes. In our work, we adopt the prototype refining method based on the subspace projection distance.

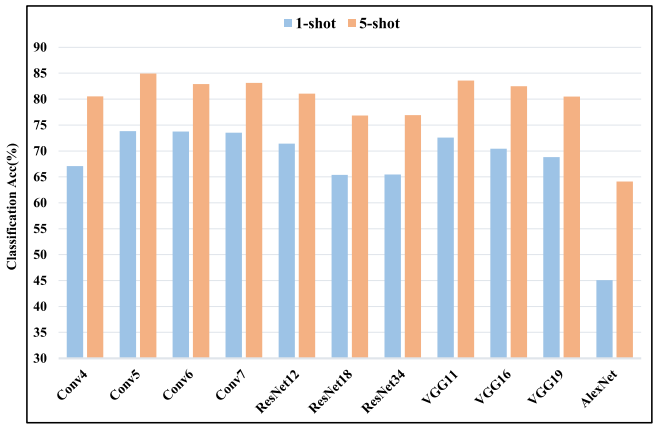

Fig. 9. Performance comparison of different feature extraction networks on the NWPU-RESISC45 dataset.

The probabilities that query samples belong to each category are calculated according to the projected distance of query samples on original class subspaces. To compare these two prototype refining methods, we make experiments on NWPU-RESISC45, UC Merced, and AID datasets, respectively. Table VIII presents the experimental results under one-shot and five-shot scenarios.

As can be seen from Table VIII, in the process of constructing an updated subspace classifier, adopting the prototype refining method based on the subspace projection distance achieves better results on the final classification accuracy. Through analyzing the experimental results, we can draw the following conclusions. First, the refined prototype obtained by our method is able to get an updated subspace classifier with better classification performance. Second, the performance of the updated subspace classifier depends on the accuracy of class prototypes. Compared with the refining method based on prototype distance, the refined prototype obtained by our method can better represent the overall situation of one class. This precisely reflects that the class prediction of the original subspace classifier on the query set is closer to a real situation, and the classification reliability is higher. The subspace classifier is more suitable for FSRSSC than the prototype classifier.

4) Impact of Feature Extraction Network: In this section, we explore the impact of different feature extraction networks on few-shot classification performance in terms of both network complexity and network depth. We select 11 models as the backbone to perform feature extraction on the images. These models can be broadly classified into four groups: simple networks (Conv4, Conv5, Conv6, and Conv7), ResNet-based networks (ResNet12, ResNet18, and ResNet34), VGG-based networks (VGG11, VGG16, and VGG19) and other networks (AlexNet). Fig. 9 shows the experimental results on the more challenging NWPU-RESISC45 dataset. First, by analyzing the four groups of models, we can find that the simple networks present better classification performance compared to the complex network and are more suitable for the metric-based few-shot classification methods. Second, by analyzing the models in each group individually, we can see that the few-shot classification accuracy is not improved with the increase of network depth, and even shows a decreasing trend. This is because few-shot learning-based methods use a small number of labeled samples for training, which generates

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TABLE VII

ABLATION STUDY OF UPDATED SUBSPACE CLASSIFIER ON NWPU-RESISC45, UC MERCED, AND AID DATASETS

Setting	NWPU-RESISC45		UC Merced		AID	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
w/o updated subspace classifier	64.94±0.88	81.56±0.53	54.90±0.76	74.20±0.48	60.29±0.79	79.76±0.49
w/ updated subspace classifier	**73.84±0.91**	**84.96±0.46**	**62.20±0.84**	**79.44±0.47**	**66.00±0.86**	**83.36±0.46**

TABLE VIII

ABLATION STUDY OF PROTOTYPE REFINING METHODS ON NWPU-RESISC45, UC MERCED, AND AID DATASETS

Prototype Refining Methods	NWPU-RESISC45		UC Merced		AID	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
prototype distance-based	71.88±0.93	84.23±0.46	61.16±0.91	77.11±0.52	64.12±0.86	82.47±0.49
subspace projection distance-based	**73.84±0.91**	**84.96±0.46**	**62.20±0.84**	**79.44±0.47**	**66.00±0.86**	**83.36±0.46**

TABLE IX

COMPARISON OF CLASSIFICATION PERFORMANCE OF OPTIMAL MODELS TRAINED WITH DIFFERENT LOSS FUNCTIONS ON NWPU-RESISC45, UC MERCED, AND AID DATASETS

Loss Function	NWPU-RESISC45		UC Merced		AID	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
Cross Entropy Loss	72.74±0.89	84.58±0.50	59.24±0.81	77.43±0.48	64.79±0.85	83.01±0.46
Our loss	**73.84±0.91**	**84.96±0.46**	**62.20±0.84**	**79.44±0.47**	**66.00±0.86**	**83.36±0.46**

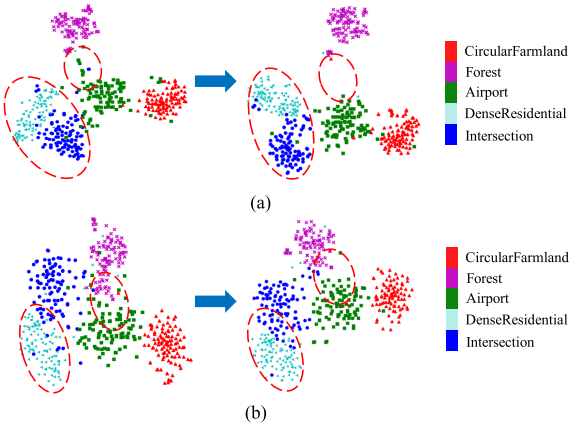

Fig. 10. t-SNE visualization results of the optimal models trained with different loss functions on the NWPU-RESISC45 dataset. (a) t-SNE visualization results in the case of one-shot, 72.74% acc with the cross-entropy loss on the left and 73.84% acc with our loss function on the right. (b) t-SNE visualization results in the case of five-shot, 84.58% acc with the cross-entropy loss on the left and 84.96% acc with our loss function on the right.

meta-overfitting [56] in the face of complex and deep networks. Therefore, Conv5 model is chosen as our feature extraction network in this work.

5) Effect of Loss Functions: In our work, we add the interclass constraint and intraclass constraint to mitigate the effect of higher interclass similarity and larger intraclass variance of remote sensing scene images on classification. Next, we verify whether the setting of our loss function is helpful to obtain a better classification model. We conduct a set of comparison experiments on NWPU-RESISC45, UC Merced, and AID datasets. The network architecture of all experiments remains the same, but different loss functions are applied, that is, one group adopts cross entropy-loss, and the other group adopts our loss function. The experimental results are shown in Table IX. It is easy to see that, no matter the case of one-shot or five-shot, the model trained by our loss function has higher classification accuracy on test tasks. Therefore, for FSRSSC tasks, our loss function has the ability to maximize the interclass distance and intraclass similarity to get a better classification model.

In order to more intuitively reflect the ability of the optimal training model obtained by different loss functions, taking the NWPU-RESISC45 dataset as an example, we carry out t-distributed stochastic neighbor embedding (t-SNE) visualization [57] on the test set of NWPU-RESISC45 dataset. As with the task setup, five categories are randomly selected from all test categories. And then, we randomly select 100 images from each category for t-SNE visualization to obtain their mapping points on the 2-D plane. As can be seen from Fig. 10, the model obtained by our loss function pays more attention to the gathering of images in the same class, and the distance between classes is more obvious.

V. CONCLUSION

In this work, we propose an FSRSSC method based on subspace network. We build a subspace classifier that is more suitable for remote sensing scene images. It takes the feature subspace as a class metric. The feature subspace can effectively represent the commonalities of a category, and mitigate the negative impact of category-irrelevant objects and other background information on the performance of the classifier. In particular, we add the information of the query set to update the original subspace classifier for an updated subspace classifier. The updated subspace classifier has more accurate class subspaces and better classification performance. In addition, we propose an interclass constraint and an intraclass constraint to mitigate the classification confusion caused by higher interclass similarity and larger intraclass variance of remote sensing scene images. They can make the images of different classes as far away as possible and the images of the same class as closely clustered as possible in the feature space. Experimental results on three public remote sensing scene datasets verify the effectiveness of our method.

In future work, we will focus on how to exploit the deeper network more efficiently without meta overfitting to extract distinguishing features and how to increase the confidence of estimated probabilities of query set, aiming to further improve the accuracy of refined prototypes and the classification performance of subspace classifier.

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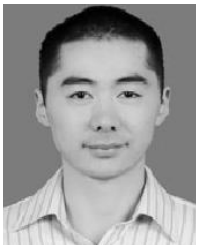

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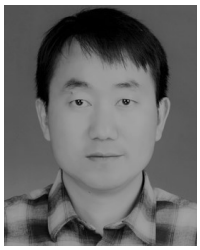

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